What is a design pattern..?

A design pattern is a reusable, proven solution to a commonly occuring problem in a software design.

* It is not actual code that you can copy & paste directly..

Ex If an architect wants to build a staircase, they do not invent a new staircase every time. They use a known design already works well..

⇓

In the same way, software developers use design pattern many coding problems repeat across projects"

Instead, it is an template or idea that tells you:

- what problem exists
- why that problem happens
- what structure can solve it
- when to use that solution
- what advantages & disadvantages it has..

** "A design pattern is a standard way to solve a recurring software design problem"

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Why design patterns are Used:-

Design patterns are used because they help developers while building a Software service/product

- write cleaner code
- Reuse proven solutions
- Make code easier to understand
- Reduce bugs
- Improve communication between developers
- Make Systems easier to maintain & extend
- Avoid reinventing the wheel..

Real life Example:-

Imagine you are building houses:-

Different houses may need:-

+ Doors
+ windows
+ Roofs
+ Staircase

You do not invent these things from scratch every time. You use existing designs that are already tested & works well...

So, Design patterns are similar in

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Note:

A design pattern is not :-

- A programming language
- A library
- A frame work
- Ready-made code

It is only a solution idea or structure..

You can implement same pattern in many programming language like Java, Python, & c++, Java Script, etc..

Part of Design pattern

Name - Problem - Solution - Consequences

So..

A design pattern is a reusable, standard solution to common software design problem that helps developers to write better, cleaner, & more maintainable code...

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